# Geometric Cabinet Command Cards

Take the square from the squares tray. Remove the square from its frame. Check the geometry cabinet to see how many shapes fit into the square frame.

Draw a red line on a piece of paper that is 10 cm long. Check insets from different drawers to see how many figures are 10 cm long.

Fold a paper lengthwise into five equal columns. Label each column according to their number of sides: 1 side, 2 sides, 3 sides, 4 sides, 5 or more sides. List the shapes in the correct column.

Take out the polygon drawer from the cabinet. Rotate each inset and have the children imagine what it would be like to have that shape as wheels on a bike. Predict which ride would be the roughest and which would be the smoothest.

Spread out triangles from the triangle drawer. Place three labels at the top of a rug: scalene isosceles equilateral Place the triangles in the correct row. If it belongs in more than one row, place it between the rows.

Take the circle drawer to your rug and remove all the circles. On a piece of paper, trace the circles to make a random design of overlapping, multi-colored circles.

Geo Cabinet Cards cont'd

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| Make two labels: one that says "Parallelogram" and one that says "Quadrilateral". Take the square tray and the quadrilateral tray to the rug and sort the shapes. How many of the shapes are both parallelograms and quadrilaterals? |
| Draw all the triangles from the triangle drawer and measure each side. Then add up the sides of each triangle to determine the perimeter of each. Which triangle has the greatest perimeter and which has the smallest perimeter? |
| Make two labels: one that says "Three Sided Figures", and one that says "Four Sided Figures". Take the triangle drawer and the quadrilateral drawer to your rug and group the shapes according to the labels. |
| Bring the polygon drawer to your rug, with a box of bead bars from the math area. Make stars by placing beads along the sides of each shape (starting with the bead that best fits the side of each polygon), and then continuing down to one bead. |
| Bring the circle tray to your rug. On one piece of paper, trace each circle inside the other, starting with the largest and ending with the smallest. You have made concentric circles—they share the same axis (center) but have different radii. |
| Take the polygon tray to your rug. For each polygon, count the number of sides and the number of angles. What do you discover? |
| Bring the tray of varied, curvilinear figures to your rug. Choose one shape and trace it. Using colored pencils, fill in your shape with straight lines that are both parallel and intersecting. Label the shape. |

# Constructive Triangle Command Cards

Take the triangle box back to your rug. Group the same-colored triangles, connecting along black guidelines. Compare the gray equilateral triangle to the other triangles. What do you notice?

With the contents of the triangle box, group the same-colored triangles, connecting along sides as to not match the black guidelines. What shapes can you make? Draw them on a piece of paper.

With the contents of the triangle box, take one triangle of each color. Label them: equilateral, right-angled scalene, obtuse isosceles, and equilateral. Which two triangles have the same shapes but different size?

With the contents of the triangle box, use fraction labels to compare each green, yellow, and red triangular piece, as compared to the gray equilateral triangle. Draw and label what you find.

With the contents of the large hexagon box, arrange pieces of the same color along black guidelines. Label the shapes formed: hexagon, equilateral triangle, rhombus, and parallelogram. Draw them on a piece of paper.

Use a ruler and protractor to measure the shapes in the large hexagon box. Draw congruent figures on paper and label them.

Command Cards for the Constructive Triangle Boxes cont'd

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| With the contents of the small hexagon box, arrange same-colored triangles along black guidelines. Label the shapes formed: hexagon, trapezoid, equilateral triangle, (4) rhombus. |
| With the contents of the small hexagon box, group the congruent triangles. Are there any similar triangles? |
| With the contents of the small hexagon box, compare the values of all the same color shapes. Label the figure with appropriate fraction cards. |
| Use the contents of the triangle box, small hexagon box, and large hexagon box to experiment with different constructions. |
| Use the contents of the triangle box, small hexagon box, and large hexagon box to make comparisons among the shapes. Which are congruent? Which are similar? |
| With the contents of the rectangle box, arrange pieces of the same color along black guidelines. Label the shapes formed: square, parallelograms, rectangle, rhombus, and trapezoid. Draw them on a piece of paper. |
| Bring the constructive triangle boxes to the rug. Use all the triangles to construct one parallelogram. |

# Fundamental Concepts Commands

A1

Look around you. How many points can you find? Draw and label some of the points you see.

A1

Look around the room. How many lines do you see? Make a list of the lines you see, and draw at least one of them.

A1

Look around you. How many surfaces do you see? Draw and label some of the surfaces that you see.

A1

Look around the room. See how many solids you can find. Make a list of the solids you find, and draw a picture of one of them.

A1

Take a tray and put the following Golden Beads on it: 10 unit beads, 10 ten bars, and 10 hundred squares. Use the units to make a line, the ten bars to make a surface, and the hundred squares to make a solid.

A1

Take some clay or play-doh. Use a tiny bit to make a point. Roll some out to make a line. Flatten some to make a surface. Mold some to make a solid. If you can, let it dry and then paint your creations.

# Types of Lines Command Cards

B1

Look around the room and find some examples of straight lines. Using a blue pencil, draw and label some of them on your paper.

B1

Look around the room and find some examples of curved lines. Using a red pencil, draw and label some of them on your paper.

B1

Using different colored pencils, make a design on a piece of paper that is a mixture of straight lines and curved lines. Be creative!

B1

Fold a piece of paper in half and write these two headings at the top: straight lines curved lines
List things you see in the room under the right heading.

B1

Place labels on a rug that say: straight line curved line
Find five shapes from the geometry cabinet that fit under each label and place them correctly on the rug.

B1

Fold a piece of paper in half and write these two headings at the top: straight lines curved lines
Think of a specific place (the beach, a park, a circus) and write things you would find there under the correct heading.

Line Command Cards Cont'd

| |
|---|
| B2<br>Draw a line on a piece of paper. Make one end a dot and the other an arrow. Label the parts: ray, end point, origin. |
| B2<br>Draw a line segment with a green pencil. Mark the end points with red dots. Draw a ray with a blue pencil. Mark the origin with a red dot and show how it extends with a black arrow. |
| B3<br>Make three columns on a piece of paper: horizontal, vertical, and oblique. List as many real-life examples of each kind of line as you can. |
| B3<br>On a piece of paper, draw a horizontal line, a vertical line, and an oblique line. How can you show that the lines go on forever? Do this to the lines you drew. |
| B3<br>Think of a picture you would like to draw with straight lines. When you are drawing your picture, use the following color code: red for horizontal lines, blue for vertical lines, and yellow for oblique lines. |
| B3<br>Bring the stick box to the rug and put out some sticks. Use red to make horizontal lines, blue for vertical lines, and yellow for oblique lines. |
| B5<br>Look around you and find at least one example of parallel lines. Draw and label the parallel lines on your paper. |

Line Command Cards Cont'd

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|---|
| B5<br>Using a ruler, draw a pair of divergent lines on a piece of paper. Use arrows at the ends of the lines to indicate which direction they are going. |
| B5<br>Using a ruler, draw a pair of convergent lines on a piece of paper. Use arrows at the ends of the lines to indicate which direction they are going. |
| B5<br>Pick one point on a piece of paper. Draw as many concurrent lines through that one point as you can. Use a different color for each line if you'd like. |
| B5<br>Take a piece of paper and cover it with lines that are going in all directions. Now count: how many horizontal lines did you make? Vertical? Oblique? Intersecting? Parallel? Convergent? Divergent? Perpendicular? |
| B5<br>Using the quadrilateral drawer of the Geometry Cabinet, count up how many different kinds of lines were used to make the shapes you see. Trace the shapes and label the lines used to make them. |
| B5<br>Take two sharp pencils, one red and one black, and bind them together with a rubber band. Resting the tip of one pencil on the edge of a ruler, draw a line with both pencils. Check |
| B5<br>Take two sharp pencils, one red and one black, and bind them together with a rubber band. On a sheet of paper, draw a curved path with both pencils. Check the distance between |

Line Command Cards Cont'd

B5

Draw an oblique line; then draw one parallel to it.
Draw a vertical line; then draw one parallel to it.
Draw a horizontal line; then, a draw one parallel to it.

B5

Use your geometry sticks to form convergent lines. Add black arrows to show the direction the lines are moving. Now, form divergent lines and add arrows indicate direction. Place the correct label under each of your formations.

B5

Form convergent lines with two sticks and add arrows to indicate direction. How can you make them divergent without touching the sticks?

*Move the arrows!*

B5

Use your ruler and arrows to show parallel, divergent and convergent lines. Use red arrows for parallel lines, blue arrows for convergent lines, and yellow arrows for divergent lines. Label your illustrations.

B5

Take out the quadrilateral drawer. Observe the types of lines used to form the figures. Which lines were used the most? Which line was used once? Using black arrows, can you show divergent and convergent lines on the trapezoid.

# Angles Command Cards

C1

Use the red geometry sticks to form an angle. Label it. Make three other angles using a different color for each angle. What differences do you note?

*The size of each angle is different.*

C2

Use the geometry sticks, arrows, and tacks to form an angle. Use a red tack for the vertex, yellow sticks for the sides, and a blue tack for the size.

C2

Draw an angle on your paper using red to highlight the angle size. Use the following code to indicate the angle parts:
1 = sides, 2 = size, 3 = vertex, 4 = angle

C3

Use the stick box to make a green acute angle, a red obtuse angle, a blue right angle and a brown reflex angle. Place a label under each illustration.

C3

Make a large circle on your paper. Indicate the vertex with a red dot. Make a green vertical line through the vertex. Then, draw an red acute angle, a blue straight angle, and a brown whole angle. Label them.

C3

Using a geo board and rubber bands, create an angle. Then, have a friend change the rubber bands to make a different angle. Switch and let your friend create an angle first, then you change it.

Angles Commands Cont'd

| | |
|---|---|
| Put the angle labels on the rug at one end. Place the geometry sticks at the other. Form an angle, and then find the correct label for it. | C3 |
| Select the triangle drawer and bring it to your rug. Trace each triangle on your paper, then color the angles using this code: red for acute, blue for right angles, and yellow for obtuse. | C3 |
| Place these labels at the top of the rug: obtuse, right, acute Take out the rectangle box and set each triangle under the correct heading. | C3 |
| Make two angles with one side the same color in each angle. Combine the angles, removing one of the rays that has the same color so that they share a ray. Label them "adjacent". | C4 |
| Form two angles with the stick box, using the same colors for each angle. Join the angles at the vertex. Exchange each of the two sides for one longer side. Label them "vertical". | C4 |
| Create a right angle with two sticks from the stick box. Add another stick to create a complementary angle. Then create a supplementary angle with two sticks. Draw them on paper and label them. | C5 |
| Use your ruler to draw a set of complementary angles and a set of supplementary angles. Color any acute angles red and any obtuse angles blue. Label them. | C5 |

# Closed Figures, Polygons, and Closed Curves

D1

Sort the shapes from the Geometric Cabinet into two sets: polygons and simple closed curves. Trace and label some of the shapes.

D1

Use the Geometric Stick Box to construct a simple closed curve. Draw and label it. Color the inside of the closed curve orange.

D1

Use the Geometric Stick Box to construct a polygon. Draw and label it. Color the inside of the polygon green.

D2

Find all of the triangles and all of the quadrilaterals in the Geometric Cabinet. Trace and label some of them.

D2

Use the Geometric Stick Box to construct three different triangles and three different quadrilaterals. Draw and label them.

D3

Locate, trace, and label all the circles, ovals, and ellipses in the Geometric Cabinet.

# Triangles

E1

Examine the Geometric Solids. Which have triangular faces? What kinds of triangles are they?

E1

Sort the Detective Triangle Game triangles into three groups:
**equilateral** **isosceles** **scalene**

E1

Use the Geometric Stick Box to construct an equilateral triangle, an isosceles triangle, and a scalene triangle. Draw and label them.

E1

Find and draw all the triangles in the Geometric Cabinet. Label each equilateral, isosceles, or scalene.

E2

Draw one example of each type: acute triangle, obtuse triangle, and right-angled triangle. Measure and mark the angles of each. Find the sum of angles for each and mark it down.

E2

Sort the Detective Triangle Game triangles into three groups:
**acute** **obtuse** **right-angled**

# Quadrilaterals and Circles

F1

Draw a square with a pencil and ruler. Cut out the square. Cut it up so each of the angles is separate. Put the angles together and glue them on another paper. What did you make?

F1

Construct a rhombus with a pencil, ruler, and protractor. Cut out the rhombus. Cut it up so each of the angles is separate. Glue the angles together on paper. What did you make?

F1

Use the Stick Box to construct the following shapes: trapezoid, parallelogram, rhombus, rectangle, square, and common quadrilateral. Draw and label them.

H1

Use a compass to make a design of circles that may or may not intersect. Color them beautifully.

H1

Find five circular objects in your home or classroom (or even better, go outside to look!) Sketch them.

H1

Challenge: Inscribe each of the triangles in the Geometric Cabinet inside its own circle. (To inscribe means that a shape fits inside another with just the points touching).

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# Stick Box Command Cards

Use the stick box to construct different kinds of angles: right angle, straight angle, acute angle, obtuse angle, and reflex angle. Draw and label each angle.

Use the stick box to construct 5 different quadrilaterals. Draw and label them on a piece of paper.

Use the stick box to construct 3 pairs of supplementary angles. Draw and label them. Are the angles acute, obtuse, or right?

Bring the stick box to your rug. Use the sticks to create different combinations of lines, including: vertical, horizontal, oblique, intersecting, perpendicular, concurrent, and bisecting.

Use the stick box to construct adjacent line segments. Use two different colors to draw them. Label your drawing.

Use the stick box to make different triangles according to sides: scalene, equilateral, and isosceles. Draw and label them.

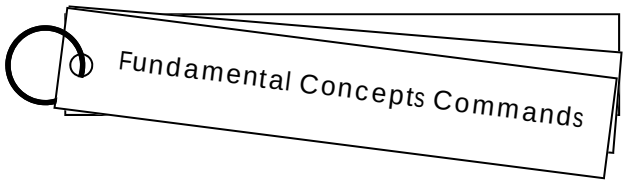
**Making the Command Card Sets:**

Print on cardstock, laminate, and cut cards apart. It would be nice to print them on different colors for each set, but you can also print all the pages on light blue (for geometry) and then laminate and cut cards apart.

**Using the Command Card Sets:**

These command cards are intended for children who have completed the cards or folders for that category. They are extensions for a child who is ready to be challenged.

The cards are formatted so that you can punch a hole (with a hole punch) at the far left of each card, after cutting them apart. Then the cards can be grouped by category.

The cards should be placed near the other Geometry materials for ease of use (setting them in a basket is nice).

A good way to hold them together is with a metal ring (found at office supply stores). Or, you can use string to tie them together. They would look like this (with the title card on top):